Supplement to NeST-Com: Fixed-Budget Next-Scale Token Communication

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Complete experimental results and implementation details. Numerical tables use the same frozen CSV sources as the main manuscript.

S1 System settings and interpretation

This supplement accompanies the token-wise configuration of NeST-Com. The main method is the original frozen tokenizer, VAR-d16, adapted D_c, and argmax receiver with a finer legal endpoint set. Historical Quarter, Kodak, coarse-only and Ada-argmax results remain in their original versioned deliveries; they are not relabelled as new token-wise tests. No model training or parameter scan was performed for this evaluation.

S1.1 Policies and actual frame accounting

The header contains a 12-bit profile and CRC16, convolutionally protected in 68 symbols. The body pays a 13-bit length and CRC16; Ada additionally pays its 10-bit predicted class. Independent termination produces actual arithmetic streams. Main NeST and WHOLE targets and MCSs are shown below. Both budgets use $E_s = 2$, and the smaller frame has 444 body symbols rather than 956. NeST retains the existing targets and MCSs while using integer-token endpoints. Family-specific calibration and target choices are therefore distinct from the matched-ceiling controls.

N	SNR	NeST target	WHOLE target	Modulation	Rate	Source cap.
1024	4	(7,25)	(7,0)	4-QAM	2/3	1245
1024	7	(10,0)	(10,0)	16-QAM	1/2	1883
1024	10	(8,126)	(9,0)	16-QAM	2/3	2520
1024	19	(10,0)	(10,0)	64-QAM	5/6	4751
512	4	(5,27)	(6,0)	4-QAM	2/3	563
512	10	(7,0)	(7,0)	16-QAM	2/3	1155
512	19	(8,84)	(8,0)	64-QAM	5/6	2191

QPSK is listed as 4-QAM. Source capacity is the body information length minus its 29-bit length/CRC overhead. A full 256-token Ada representation can leave unused capacity; that is not a fitting failure. The $N = 1024$ no-fit branch raises an error (none observed), whereas $N = 512$ has a paid NULL branch. Receiver failures and NULL retain the fixed gray image and the complete statistical denominator.

The original 4/10/19-dB policies use an initial 100-image screening stage with one channel realization per image and final cal1000 evaluation with three realizations per image. The 7-dB extension uses the previously frozen finite 100-image, five-MCS calibration. No test-set score reselects a policy here. DINOv2 cosine was the calibration objective. The DISTS, DreamSim and CLIP feature criteria are related to that objective and do not constitute independent human perceptual judgments.

S1.2 Models, metrics, and source identity

The decoder adaptation uses 20 000 ImageNet training images, MSE plus 0.1 LPIPS, and a 50/25/25 mixture of continuous, fully quantized, and interpolated prefix-completed latents. The final category interpolates continuous latents with prior completions conditioned on eight received source scales. Tokenizer, VAR prior and codebook are frozen. ARPC also uses next-scale representations, and must not be treated as a raster-order control. Ada is a predicted-class fixed-frame adaptation, with ConvNeXt category prediction, paid category bits, original conditional entropy CDFs, and CFG4/T1/top-k2000/top-p1 tail generation. It does not reproduce the long-term queue or ARQ controller.

DISTS uses the official revision `1267d8cb626c98706db3697422701c56a85ebf2e`, learned alpha/beta weights and VGG16 ImageNet weights, in evaluation FP32 without AMP. Its input is the native 256×256 float32 RGB array without extra cropping, resizing, alignment or PNG quantization. DreamSim uses the frozen ensemble and 224×224 floating-point bicubic-antialias preprocessing; LPIPS-Alex, PSNR and CLIP-I retain the established routes. The previously missing

external 7-dB CLIP values were scored separately in the matching Torch2.10 route, with no cross-version feature-cache mixing.

Full source IDs, native pixel hashes, preprocessing, actual receiver inputs, model hashes and numerical routes control every reuse decision. Common500 means first average three channel events per source. The original $10\,000 \times 500$ source-bootstrap bank supplies pointwise percentile intervals and direct paired differences. The control100 bank has its own fixed source order and $10\,000 \times 100$ indices; it is not a slice of the 500-source bank. No multiplicity-adjusted simultaneous claim is made. The exact hashes and orders accompany the CSVs.

S2 Complete quality results

S2.1 Five-metric source means

All means retain failures. Pointwise intervals and direct paired intervals are provided in the full-precision CSVs; the compact table reports means.

N	SNR	Method	PSNR↑	LPIPS↓	DreamSim↓	CLIP-I↑	DISTS↓
512	4	Ada	11.318	0.5542	0.3847	0.7892	0.2996
512	4	WHOLE	15.107	0.3951	0.3247	0.7469	0.2580
512	4	NeST	15.611	0.3565	0.2740	0.7691	0.2363
512	4	Quarter	15.499	0.3652	0.2819	0.7657	0.2406
512	10	Ada	13.165	0.4110	0.2582	0.8257	0.2367
512	10	WHOLE	16.821	0.2881	0.2049	0.8074	0.2030
512	10	NeST	17.413	0.2650	0.1837	0.8211	0.1922
512	10	Quarter	17.220	0.2726	0.1922	0.8150	0.1958
512	19	Ada	17.690	0.2146	0.1078	0.8723	0.1496
512	19	WHOLE	18.638	0.2175	0.1458	0.8484	0.1718
512	19	NeST	19.632	0.1832	0.1150	0.8689	0.1554
512	19	Quarter	19.430	0.1901	0.1217	0.8640	0.1587
1024	4	ARPC	14.184	0.5124	0.5010	0.6442	0.3436
1024	4	Ada	13.332	0.3952	0.2431	0.8280	0.2300
1024	4	BPG	20.743	0.7255	0.5737	0.5859	0.4022
1024	4	Latent	19.658	0.2188	0.2317	0.7408	0.2218
1024	4	Swin	21.671	0.5297	0.4359	0.6267	0.3488
1024	4	WHOLE	16.832	0.2878	0.2046	0.8077	0.2028
1024	4	NeST	17.654	0.2549	0.1743	0.8271	0.1882
1024	4	Quarter	17.448	0.2633	0.1839	0.8206	0.1916
1024	7	ARPC	14.759	0.4809	0.4751	0.6508	0.3287
1024	7	Ada	15.905	0.2698	0.1408	0.8592	0.1766
1024	7	BPG	21.717	0.6500	0.4788	0.6150	0.3676
1024	7	Latent	20.369	0.1893	0.1912	0.7717	0.2070
1024	7	Swin	22.475	0.4933	0.3887	0.6505	0.3336
1024	7	WHOLE	18.528	0.2228	0.1507	0.8460	0.1756
1024	7	NeST	18.981	0.2085	0.1399	0.8539	0.1696
1024	7	Quarter	18.850	0.2114	0.1420	0.8507	0.1699
1024	10	ARPC	16.471	0.3413	0.3062	0.7134	0.2278
1024	10	Ada	19.559	0.1674	0.0871	0.8841	0.1275
1024	10	BPG	22.543	0.5894	0.4165	0.6350	0.3409
1024	10	Latent	20.803	0.1733	0.1712	0.7892	0.1986
1024	10	Swin	23.026	0.4687	0.3564	0.6691	0.3224
1024	10	WHOLE	19.313	0.1987	0.1317	0.8577	0.1621
1024	10	NeST	20.194	0.1658	0.1048	0.8769	0.1481
1024	10	Quarter	20.014	0.1715	0.1077	0.8731	0.1504
1024	19	ARPC	18.199	0.2503	0.1952	0.7880	0.1812
1024	19	Ada	20.938	0.1376	0.0762	0.8925	0.1136
1024	19	BPG	24.230	0.4584	0.2964	0.6843	0.2920
1024	19	Latent	21.245	0.1585	0.1546	0.8041	0.1911

N	SNR	Method	PSNR↑	LPIPS↓	DreamSim↓	CLIP-I↑	DISTS↓
1024	19	Swin	23.698	0.4400	0.3179	0.6966	0.3081
1024	19	WHOLE	21.844	0.1234	0.0815	0.8969	0.1274
1024	19	NeST	22.334	0.1151	0.0772	0.9015	0.1226
1024	19	Quarter	22.164	0.1182	0.0785	0.9005	0.1243

S2.2 Direct paired differences

NeST minus the stated reference. Positive PSNR/CLIP-I and negative distance differences favor NeST. A confidence interval spanning zero does not establish a direction.

N	SNR	Reference	Metric	Difference [95% CI]
512	4	Quarter	PSNR	+0.11210 [0.06798, 0.16080]
512	4	Quarter	LPIPS	-0.00863 [-0.01104, -0.00621]
512	4	Quarter	DreamSim	-0.00782 [-0.01132, -0.00447]
512	4	Quarter	CLIP-I	+0.00345 [-0.00043, 0.00738]
512	4	Quarter	DISTS	-0.00433 [-0.00627, -0.00247]
512	10	Quarter	PSNR	+0.19304 [0.16224, 0.22489]
512	10	Quarter	LPIPS	-0.00759 [-0.00914, -0.00622]
512	10	Quarter	DreamSim	-0.00842 [-0.01061, -0.00630]
512	10	Quarter	CLIP-I	+0.00611 [0.00330, 0.00898]
512	10	Quarter	DISTS	-0.00367 [-0.00476, -0.00270]
512	19	Quarter	PSNR	+0.20191 [0.17590, 0.22869]
512	19	Quarter	LPIPS	-0.00681 [-0.00771, -0.00595]
512	19	Quarter	DreamSim	-0.00669 [-0.00828, -0.00521]
512	19	Quarter	CLIP-I	+0.00489 [0.00239, 0.00738]
512	19	Quarter	DISTS	-0.00326 [-0.00391, -0.00263]
1024	4	Quarter	PSNR	+0.20601 [0.17556, 0.23714]
1024	4	Quarter	LPIPS	-0.00831 [-0.00935, -0.00728]
1024	4	Quarter	DreamSim	-0.00960 [-0.01196, -0.00732]
1024	4	Quarter	CLIP-I	+0.00648 [0.00334, 0.00973]
1024	4	Quarter	DISTS	-0.00343 [-0.00420, -0.00268]
1024	7	Quarter	PSNR	+0.13138 [0.07917, 0.18093]
1024	7	Quarter	LPIPS	-0.00292 [-0.00689, 0.00135]
1024	7	Quarter	DreamSim	-0.00210 [-0.00668, 0.00274]
1024	7	Quarter	CLIP-I	+0.00318 [-0.00002, 0.00639]
1024	7	Quarter	DISTS	-0.00024 [-0.00368, 0.00344]
1024	10	Quarter	PSNR	+0.18002 [0.15807, 0.20281]
1024	10	Quarter	LPIPS	-0.00567 [-0.00635, -0.00502]
1024	10	Quarter	DreamSim	-0.00292 [-0.00379, -0.00204]
1024	10	Quarter	CLIP-I	+0.00379 [0.00178, 0.00581]
1024	10	Quarter	DISTS	-0.00235 [-0.00286, -0.00185]
1024	19	Quarter	PSNR	+0.17033 [0.15600, 0.18558]
1024	19	Quarter	LPIPS	-0.00304 [-0.00332, -0.00277]
1024	19	Quarter	DreamSim	-0.00127 [-0.00164, -0.00090]
1024	19	Quarter	CLIP-I	+0.00100 [-0.00007, 0.00209]
1024	19	Quarter	DISTS	-0.00166 [-0.00194, -0.00138]
512	4	WHOLE	PSNR	+0.50427 [0.43976, 0.57335]
512	4	WHOLE	LPIPS	-0.03858 [-0.04275, -0.03445]
512	4	WHOLE	DreamSim	-0.05062 [-0.05665, -0.04454]
512	4	WHOLE	CLIP-I	+0.02219 [0.01714, 0.02740]
512	4	WHOLE	DISTS	-0.02171 [-0.02449, -0.01904]
512	10	WHOLE	PSNR	+0.59161 [0.54369, 0.64265]
512	10	WHOLE	LPIPS	-0.02307 [-0.02452, -0.02161]
512	10	WHOLE	DreamSim	-0.02112 [-0.02363, -0.01865]
512	10	WHOLE	CLIP-I	+0.01365 [0.01009, 0.01718]

N	SNR	Reference	Metric	Difference [95% CI]
512	10	WHOLE	DISTS	-0.01082 [-0.01180, -0.00986]
512	19	WHOLE	PSNR	+0.99391 [0.93976, 1.04841]
512	19	WHOLE	LPIPS	-0.03430 [-0.03577, -0.03284]
512	19	WHOLE	DreamSim	-0.03081 [-0.03328, -0.02843]
512	19	WHOLE	CLIP-I	+0.02054 [0.01696, 0.02425]
512	19	WHOLE	DISTS	-0.01644 [-0.01747, -0.01542]
1024	4	WHOLE	PSNR	+0.82204 [0.76554, 0.88112]
1024	4	WHOLE	LPIPS	-0.03289 [-0.03452, -0.03128]
1024	4	WHOLE	DreamSim	-0.03033 [-0.03335, -0.02731]
1024	4	WHOLE	CLIP-I	+0.01942 [0.01560, 0.02316]
1024	4	WHOLE	DISTS	-0.01463 [-0.01571, -0.01357]
1024	7	WHOLE	PSNR	+0.45308 [0.39299, 0.51258]
1024	7	WHOLE	LPIPS	-0.01432 [-0.01880, -0.00967]
1024	7	WHOLE	DreamSim	-0.01084 [-0.01567, -0.00566]
1024	7	WHOLE	CLIP-I	+0.00793 [0.00439, 0.01153]
1024	7	WHOLE	DISTS	-0.00603 [-0.00966, -0.00207]
1024	10	WHOLE	PSNR	+0.88117 [0.80754, 0.95550]
1024	10	WHOLE	LPIPS	-0.03293 [-0.03561, -0.03021]
1024	10	WHOLE	DreamSim	-0.02697 [-0.02990, -0.02408]
1024	10	WHOLE	CLIP-I	+0.01919 [0.01552, 0.02307]
1024	10	WHOLE	DISTS	-0.01406 [-0.01531, -0.01280]
1024	19	WHOLE	PSNR	+0.48975 [0.45749, 0.52298]
1024	19	WHOLE	LPIPS	-0.00824 [-0.00868, -0.00782]
1024	19	WHOLE	DreamSim	-0.00431 [-0.00508, -0.00359]
1024	19	WHOLE	CLIP-I	+0.00463 [0.00302, 0.00624]
1024	19	WHOLE	DISTS	-0.00482 [-0.00532, -0.00432]
512	4	Ada	PSNR	+4.29250 [4.09816, 4.49208]
512	4	Ada	LPIPS	-0.19761 [-0.20524, -0.19011]
512	4	Ada	DreamSim	-0.11068 [-0.12210, -0.09964]
512	4	Ada	CLIP-I	-0.02007 [-0.02843, -0.01175]
512	4	Ada	DISTS	-0.06332 [-0.06766, -0.05893]
512	10	Ada	PSNR	+4.24824 [4.07079, 4.42766]
512	10	Ada	LPIPS	-0.14597 [-0.15149, -0.14058]
512	10	Ada	DreamSim	-0.07443 [-0.08290, -0.06583]
512	10	Ada	CLIP-I	-0.00466 [-0.01139, 0.00187]
512	10	Ada	DISTS	-0.04455 [-0.04757, -0.04151]
512	19	Ada	PSNR	+1.94166 [1.77815, 2.11000]
512	19	Ada	LPIPS	-0.03133 [-0.03469, -0.02813]
512	19	Ada	DreamSim	+0.00723 [0.00374, 0.01056]
512	19	Ada	CLIP-I	-0.00339 [-0.00816, 0.00132]
512	19	Ada	DISTS	+0.00581 [0.00364, 0.00788]
1024	4	Ada	PSNR	+4.32222 [4.12839, 4.52292]
1024	4	Ada	LPIPS	-0.14026 [-0.14581, -0.13482]
1024	4	Ada	DreamSim	-0.06883 [-0.07668, -0.06109]
1024	4	Ada	CLIP-I	-0.00089 [-0.00713, 0.00524]
1024	4	Ada	DISTS	-0.04183 [-0.04498, -0.03863]
1024	7	Ada	PSNR	+3.07575 [2.88099, 3.27370]
1024	7	Ada	LPIPS	-0.06131 [-0.06696, -0.05571]
1024	7	Ada	DreamSim	-0.00093 [-0.00770, 0.00573]
1024	7	Ada	CLIP-I	-0.00527 [-0.01070, -0.00003]
1024	7	Ada	DISTS	-0.00701 [-0.01123, -0.00274]
1024	10	Ada	PSNR	+0.63508 [0.52112, 0.75347]
1024	10	Ada	LPIPS	-0.00156 [-0.00407, 0.00081]
1024	10	Ada	DreamSim	+0.01766 [0.01532, 0.01994]
1024	10	Ada	CLIP-I	-0.00721 [-0.01140, -0.00307]
1024	10	Ada	DISTS	+0.02055 [0.01885, 0.02218]
1024	19	Ada	PSNR	+1.39662 [1.33666, 1.45531]

N	SNR	Reference	Metric	Difference [95% CI]
1024	19	Ada	LPIPS	-0.02242 [-0.02342, -0.02140]
1024	19	Ada	DreamSim	+0.00095 [-0.00086, 0.00278]
1024	19	Ada	CLIP-I	+0.00903 [0.00521, 0.01290]
1024	19	Ada	DISTS	+0.00904 [0.00762, 0.01050]

S2.3 Failures and denominator

The following nonzero gray-image counts are verified from native pixel hashes and receiver statuses. Other cells have zero observed gray outputs. Zero observed failures is not zero underlying error probability. Historical noise namespaces differ between some methods and budgets, so count differences alone do not establish a causal granularity effect on frame error rate.

N	SNR	Method	Gray count	Denominator
512	4	Ada	5	1500
1024	7	Ada	7	1500
1024	7	BPG	11	1500
512	4	WHOLE	7	1500
1024	7	WHOLE	7	1500
512	4	NeST	4	1500
1024	7	NeST	13	1500
512	4	Quarter	4	1500
512	10	Quarter	1	1500
1024	7	Quarter	6	1500

S3 Matched granularity and receiver interventions

S3.1 Endpoint control

The four grids use control100, $N = 1024$, 10 dB, 16-QAM rate 2/3, a common maximum token position $(8, 126)$, and the actual shared noise vectors for repeats 6201–6203. Only legal termination points change. This WHOLE control has five permitted complete-scale endpoints; the main WHOLE system has a different target and must not be substituted. There are 1200 logical outcomes, 891 unique physical inputs, 591 newly executed physical events, and 300 exact physical reuses. Independent RX reconstruction uses 100 new and 197 reused receiver inputs.

Grid	Ends	Util. (%)	Length checks	DISTS	DreamSim
Whole	5	78.133	1.55	0.1650	0.1387
Quarter	20	95.194	4.51	0.1535	0.1143
Eighth	39	97.442	7.57	0.1516	0.1137
Token	352	99.848	125.96	0.1505	0.1115

The quality experiment reuses previously finalized streams, so its new source traversals and independent finalizations are zero. Candidate-set cardinality and descending comparisons must not be described as new network forwards. Fresh deployment computation and measured costs belong to S7. The legacy main-evaluation field named length_checks sometimes records the entire evaluated endpoint set; it is not substituted for the matched-control stopping count shown here.

Comparison	Metric	Difference [95% CI]
Quarter–Whole	PSNR	+0.65169 [0.50944, 0.80674]
Quarter–Whole	LPIPS	-0.02750 [-0.03334, -0.02195]
Quarter–Whole	DreamSim	-0.02447 [-0.03163, -0.01792]
Quarter–Whole	CLIP-I	+0.01678 [0.00815, 0.02602]

Comparison	Metric	Difference [95% CI]
Quarter–Whole	DISTS	-0.01153 [-0.01410, -0.00905]
Eighth–Whole	PSNR	+0.75730 [0.59558, 0.93119]
Eighth–Whole	LPIPS	-0.03046 [-0.03672, -0.02443]
Eighth–Whole	DreamSim	-0.02505 [-0.03221, -0.01845]
Eighth–Whole	CLIP-I	+0.01834 [0.00979, 0.02735]
Eighth–Whole	DISTS	-0.01342 [-0.01623, -0.01065]
Token–Whole	PSNR	+0.82337 [0.65877, 1.00101]
Token–Whole	LPIPS	-0.03270 [-0.03923, -0.02649]
Token–Whole	DreamSim	-0.02725 [-0.03419, -0.02071]
Token–Whole	CLIP-I	+0.02339 [0.01484, 0.03271]
Token–Whole	DISTS	-0.01454 [-0.01743, -0.01167]
Eighth–Quarter	PSNR	+0.10561 [0.06774, 0.14656]
Eighth–Quarter	LPIPS	-0.00296 [-0.00408, -0.00194]
Eighth–Quarter	DreamSim	-0.00058 [-0.00201, 0.00082]
Eighth–Quarter	CLIP-I	+0.00156 [-0.00145, 0.00456]
Eighth–Quarter	DISTS	-0.00189 [-0.00279, -0.00106]
Token–Quarter	PSNR	+0.17168 [0.12500, 0.22129]
Token–Quarter	LPIPS	-0.00520 [-0.00653, -0.00390]
Token–Quarter	DreamSim	-0.00278 [-0.00501, -0.00059]
Token–Quarter	CLIP-I	+0.00661 [0.00174, 0.01190]
Token–Quarter	DISTS	-0.00300 [-0.00411, -0.00195]
Token–Eighth	PSNR	+0.06607 [0.03357, 0.10063]
Token–Eighth	LPIPS	-0.00224 [-0.00318, -0.00131]
Token–Eighth	DreamSim	-0.00220 [-0.00422, -0.00014]
Token–Eighth	CLIP-I	+0.00505 [0.00022, 0.01051]
Token–Eighth	DISTS	-0.00112 [-0.00204, -0.00022]

S3.2 Ordered boundary-token interventions

A0 retains prior completion but omits the extra boundary-token correction and updated conditioning. B0 inserts the direct residual correction while keeping the A0 future-condition path. C0 applies both. All three arms use the same actual received prefix and retain failures. This is an ordered intervention on the full received boundary, not an additive decomposition of the tokens beyond Quarter. The complete set contains 1800 logical outcomes. It requires 394 new A0/B0 renders for 197 unique prefixes; two qualification C0 replays are pixel-identical to the official output. No new PHY events are generated.

SNR	Comparison	Metric	Difference [95% CI]
7	B0–A0	PSNR	+0.16706 [0.13852, 0.19873]
7	B0–A0	LPIPS	-0.00639 [-0.00718, -0.00558]
7	B0–A0	DreamSim	-0.00565 [-0.00774, -0.00376]
7	B0–A0	CLIP-I	-0.00061 [-0.00386, 0.00244]
7	B0–A0	DISTS	-0.00294 [-0.00383, -0.00205]
7	C0–B0	PSNR	+0.29197 [0.24095, 0.34673]
7	C0–B0	LPIPS	-0.01079 [-0.01266, -0.00895]
7	C0–B0	DreamSim	-0.00973 [-0.01344, -0.00626]
7	C0–B0	CLIP-I	+0.00860 [0.00275, 0.01511]
7	C0–B0	DISTS	-0.00456 [-0.00620, -0.00291]
7	C0–A0	PSNR	+0.45903 [0.39375, 0.52953]
7	C0–A0	LPIPS	-0.01717 [-0.01940, -0.01508]
7	C0–A0	DreamSim	-0.01538 [-0.01982, -0.01135]
7	C0–A0	CLIP-I	+0.00798 [0.00190, 0.01445]
7	C0–A0	DISTS	-0.00750 [-0.00927, -0.00576]
10	B0–A0	PSNR	+0.28681 [0.23964, 0.33531]
10	B0–A0	LPIPS	-0.01225 [-0.01448, -0.01007]
10	B0–A0	DreamSim	-0.01131 [-0.01468, -0.00828]

SNR	Comparison	Metric	Difference [95% CI]
10	B0–A0	CLIP-I	+0.00190 [-0.00310, 0.00728]
10	B0–A0	DISTS	-0.00621 [-0.00769, -0.00481]
10	C0–B0	PSNR	+0.53657 [0.40447, 0.68156]
10	C0–B0	LPIPS	-0.02045 [-0.02521, -0.01597]
10	C0–B0	DreamSim	-0.01594 [-0.02051, -0.01161]
10	C0–B0	CLIP-I	+0.02149 [0.01389, 0.02963]
10	C0–B0	DISTS	-0.00833 [-0.01022, -0.00647]
10	C0–A0	PSNR	+0.82337 [0.65877, 1.00101]
10	C0–A0	LPIPS	-0.03270 [-0.03923, -0.02649]
10	C0–A0	DreamSim	-0.02725 [-0.03419, -0.02071]
10	C0–A0	CLIP-I	+0.02339 [0.01484, 0.03271]
10	C0–A0	DISTS	-0.01454 [-0.01743, -0.01167]

At $N = 1024$, 19 dB, 1482 of 1500 NeST events have a partial final-scale boundary and 18 transmit the full representation. There is no finer scale to condition, hence B0 and C0 apply the identical final residual map. This structural check introduces no new renders, and historical Quarter A0/B0 means remain labelled Quarter.

S4 Equal information budgets

The information budget includes 12 profile bits, header CRC16, length13 and body CRC16: 57 shared bits, plus 10 class bits for Ada. The arithmetic-code limits are $B - 57$ for WHOLE/NeST and $B - 67$ for Ada. The fixed budgets are $B = 620, 1212, 1302, 1940, 2248, 2577$ bits. WHOLE/NeST share maximum $(10, 0)$; Ada permits 1–256 tokens. Actual information fields and arithmetic streams are independently parsed at RX without PHY noise. The receiver cannot inspect unsent source tokens. This comparison removes wireless and MCS effects, but retains differences among tokenizer, prior, conditioning and decoder.

WHOLE and NeST have one deterministic output per source/budget. Ada uses the three fixed generation anchors 2026103001–2026103003, with the original per-source seed mechanism; metrics, not RGBs, are averaged. There are at most 15 000 logical outputs and 9000 source-level units. These are generation repeats, not new channel realizations.

All 15 000 logical outputs were accepted by the actual source receiver; no NULL or gray output occurs in this error-free diagnostic. New reconstruction requires 195 NeST/WHOLE receiver calls and 8733 Ada calls; 4466 NeST/WHOLE and 225 Ada input identities are reused. There are 8882 newly scored native pairs; 6118 logical positions reuse exact completed metric pairs. These counts are distinct from the number of information-transfer outcomes.

At every budget all five direct NeST–WHOLE intervals favor NeST. NeST–Ada PSNR remains positive throughout. DISTS favors NeST at 620/1212/1302/1940 bits and Ada at 2248/2577 bits. DreamSim favors NeST at 620/1212/1302 bits, crosses zero at 1940 bits, and favors Ada at 2248/2577 bits. LPIPS favors NeST through 2248 bits and crosses zero at 2577 bits. CLIP-I favors Ada at 620/1940/2577 bits, with intervals crossing zero at the other budgets. These mixed results remain part of the complete-system comparison.

S4.1 Complete means and direct paired differences

Bits	Method	Metric	Mean [95% CI]
620	Ada	PSNR	11.34660 [11.16969, 11.53009]
620	Ada	LPIPS	0.54964 [0.54237, 0.55683]
620	Ada	DreamSim	0.38102 [0.36997, 0.39238]
620	Ada	CLIP-I	0.78893 [0.78097, 0.79684]
620	Ada	DISTS	0.29666 [0.29296, 0.30036]
1212	Ada	PSNR	13.06931 [12.86404, 13.27393]
1212	Ada	LPIPS	0.41291 [0.40557, 0.42011]
1212	Ada	DreamSim	0.26099 [0.25122, 0.27104]
1212	Ada	CLIP-I	0.82137 [0.81375, 0.82882]
1212	Ada	DISTS	0.23974 [0.23613, 0.24344]

Bits	Method	Metric	Mean [95% CI]
1302	Ada	PSNR	13.38734 [13.18086, 13.59338]
1302	Ada	LPIPS	0.39382 [0.38649, 0.40102]
1302	Ada	DreamSim	0.24229 [0.23279, 0.25224]
1302	Ada	CLIP-I	0.83018 [0.82262, 0.83736]
1302	Ada	DISTS	0.23054 [0.22700, 0.23419]
1940	Ada	PSNR	16.06878 [15.80180, 16.34242]
1940	Ada	LPIPS	0.26649 [0.25947, 0.27363]
1940	Ada	DreamSim	0.13658 [0.13048, 0.14283]
1940	Ada	CLIP-I	0.86146 [0.85470, 0.86805]
1940	Ada	DISTS	0.17328 [0.17004, 0.17657]
2248	Ada	PSNR	17.74405 [17.45263, 18.03827]
2248	Ada	LPIPS	0.21299 [0.20638, 0.21971]
2248	Ada	DreamSim	0.10710 [0.10246, 0.11197]
2248	Ada	CLIP-I	0.87329 [0.86679, 0.87978]
2248	Ada	DISTS	0.14885 [0.14596, 0.15185]
2577	Ada	PSNR	19.52922 [19.23907, 19.81767]
2577	Ada	LPIPS	0.16729 [0.16181, 0.17293]
2577	Ada	DreamSim	0.08696 [0.08340, 0.09072]
2577	Ada	CLIP-I	0.88382 [0.87767, 0.88999]
2577	Ada	DISTS	0.12787 [0.12544, 0.13041]
620	NeST	PSNR	15.61746 [15.38069, 15.86297]
620	NeST	LPIPS	0.35487 [0.34755, 0.36210]
620	NeST	DreamSim	0.27236 [0.26357, 0.28127]
620	NeST	CLIP-I	0.76969 [0.76034, 0.77914]
620	NeST	DISTS	0.23503 [0.23212, 0.23792]
1212	NeST	PSNR	17.41475 [17.17106, 17.67133]
1212	NeST	LPIPS	0.26496 [0.25874, 0.27117]
1212	NeST	DreamSim	0.18366 [0.17682, 0.19069]
1212	NeST	CLIP-I	0.82110 [0.81230, 0.82951]
1212	NeST	DISTS	0.19210 [0.18954, 0.19466]
1302	NeST	PSNR	17.65585 [17.40777, 17.91721]
1302	NeST	LPIPS	0.25492 [0.24880, 0.26113]
1302	NeST	DreamSim	0.17425 [0.16774, 0.18109]
1302	NeST	CLIP-I	0.82724 [0.81846, 0.83553]
1302	NeST	DISTS	0.18816 [0.18561, 0.19071]
1940	NeST	PSNR	19.04651 [18.79134, 19.31528]
1940	NeST	LPIPS	0.20235 [0.19702, 0.20764]
1940	NeST	DreamSim	0.13301 [0.12797, 0.13834]
1940	NeST	CLIP-I	0.85660 [0.84903, 0.86410]
1940	NeST	DISTS	0.16421 [0.16170, 0.16672]
2248	NeST	PSNR	19.63408 [19.36882, 19.90865]
2248	NeST	LPIPS	0.18322 [0.17818, 0.18827]
2248	NeST	DreamSim	0.11495 [0.11055, 0.11952]
2248	NeST	CLIP-I	0.86890 [0.86184, 0.87572]
2248	NeST	DISTS	0.15536 [0.15290, 0.15784]
2577	NeST	PSNR	20.19536 [19.92833, 20.47116]
2577	NeST	LPIPS	0.16577 [0.16123, 0.17038]
2577	NeST	DreamSim	0.10475 [0.10098, 0.10872]
2577	NeST	CLIP-I	0.87698 [0.87027, 0.88356]
2577	NeST	DISTS	0.14804 [0.14570, 0.15046]
620	WHOLE	PSNR	15.12224 [14.87019, 15.38250]
620	WHOLE	LPIPS	0.39246 [0.38283, 0.40199]
620	WHOLE	DreamSim	0.32180 [0.31030, 0.33372]
620	WHOLE	CLIP-I	0.74806 [0.73814, 0.75801]
620	WHOLE	DISTS	0.25563 [0.25172, 0.25960]
1212	WHOLE	PSNR	16.82131 [16.59007, 17.06388]
1212	WHOLE	LPIPS	0.28808 [0.28216, 0.29399]

Bits	Method	Metric	Mean [95% CI]
1212	WHOLE	DreamSim	0.20486 [0.19764, 0.21207]
1212	WHOLE	CLIP-I	0.80744 [0.79868, 0.81586]
1212	WHOLE	DISTS	0.20297 [0.20052, 0.20552]
1302	WHOLE	PSNR	16.83227 [16.60005, 17.07610]
1302	WHOLE	LPIPS	0.28783 [0.28187, 0.29378]
1302	WHOLE	DreamSim	0.20460 [0.19737, 0.21184]
1302	WHOLE	CLIP-I	0.80770 [0.79894, 0.81618]
1302	WHOLE	DISTS	0.20283 [0.20033, 0.20541]
1940	WHOLE	PSNR	18.55591 [18.31286, 18.81036]
1940	WHOLE	LPIPS	0.21914 [0.21398, 0.22429]
1940	WHOLE	DreamSim	0.14707 [0.14173, 0.15258]
1940	WHOLE	CLIP-I	0.84737 [0.83942, 0.85502]
1940	WHOLE	DISTS	0.17290 [0.17051, 0.17533]
2248	WHOLE	PSNR	18.63777 [18.38451, 18.90215]
2248	WHOLE	LPIPS	0.21754 [0.21220, 0.22296]
2248	WHOLE	DreamSim	0.14580 [0.14038, 0.15132]
2248	WHOLE	CLIP-I	0.84835 [0.84040, 0.85602]
2248	WHOLE	DISTS	0.17184 [0.16935, 0.17438]
2577	WHOLE	PSNR	19.31282 [19.02087, 19.60883]
2577	WHOLE	LPIPS	0.19872 [0.19218, 0.20525]
2577	WHOLE	DreamSim	0.13172 [0.12594, 0.13763]
2577	WHOLE	CLIP-I	0.85774 [0.84973, 0.86557]
2577	WHOLE	DISTS	0.16213 [0.15914, 0.16511]

Bits	Comparison	Metric	Difference [95% CI]
620	NeST–WHOLE	PSNR	+0.49521 [0.43094, 0.56242]
620	NeST–WHOLE	LPIPS	-0.03759 [-0.04141, -0.03380]
620	NeST–WHOLE	DreamSim	-0.04944 [-0.05531, -0.04362]
620	NeST–WHOLE	CLIP-I	+0.02163 [0.01652, 0.02682]
620	NeST–WHOLE	DISTS	-0.02059 [-0.02269, -0.01849]
620	NeST–Ada	PSNR	+4.27086 [4.10364, 4.44710]
620	NeST–Ada	LPIPS	-0.19476 [-0.20160, -0.18812]
620	NeST–Ada	DreamSim	-0.10866 [-0.11918, -0.09816]
620	NeST–Ada	CLIP-I	-0.01924 [-0.02717, -0.01122]
620	NeST–Ada	DISTS	-0.06163 [-0.06509, -0.05818]
620	WHOLE–Ada	PSNR	+3.77565 [3.59676, 3.96826]
620	WHOLE–Ada	LPIPS	-0.15718 [-0.16557, -0.14903]
620	WHOLE–Ada	DreamSim	-0.05922 [-0.07128, -0.04705]
620	WHOLE–Ada	CLIP-I	-0.04087 [-0.04921, -0.03263]
620	WHOLE–Ada	DISTS	-0.04103 [-0.04514, -0.03690]
1212	NeST–WHOLE	PSNR	+0.59345 [0.54574, 0.64438]
1212	NeST–WHOLE	LPIPS	-0.02311 [-0.02455, -0.02166]
1212	NeST–WHOLE	DreamSim	-0.02119 [-0.02371, -0.01875]
1212	NeST–WHOLE	CLIP-I	+0.01366 [0.01008, 0.01720]
1212	NeST–WHOLE	DISTS	-0.01088 [-0.01186, -0.00992]
1212	NeST–Ada	PSNR	+4.34544 [4.17204, 4.52565]
1212	NeST–Ada	LPIPS	-0.14794 [-0.15327, -0.14272]
1212	NeST–Ada	DreamSim	-0.07733 [-0.08526, -0.06928]
1212	NeST–Ada	CLIP-I	-0.00027 [-0.00670, 0.00590]
1212	NeST–Ada	DISTS	-0.04764 [-0.05066, -0.04462]
1212	WHOLE–Ada	PSNR	+3.75200 [3.58339, 3.92664]
1212	WHOLE–Ada	LPIPS	-0.12483 [-0.13038, -0.11947]
1212	WHOLE–Ada	DreamSim	-0.05614 [-0.06466, -0.04776]
1212	WHOLE–Ada	CLIP-I	-0.01392 [-0.02067, -0.00734]
1212	WHOLE–Ada	DISTS	-0.03677 [-0.03992, -0.03365]

Bits	Comparison	Metric	Difference [95% CI]
1302	NeST–WHOLE	PSNR	+0.82358 [0.76712, 0.88222]
1302	NeST–WHOLE	LPIPS	-0.03292 [-0.03454, -0.03131]
1302	NeST–WHOLE	DreamSim	-0.03035 [-0.03337, -0.02733]
1302	NeST–WHOLE	CLIP-I	+0.01953 [0.01574, 0.02327]
1302	NeST–WHOLE	DISTS	-0.01466 [-0.01575, -0.01359]
1302	NeST–Ada	PSNR	+4.26851 [4.10141, 4.44272]
1302	NeST–Ada	LPIPS	-0.13891 [-0.14393, -0.13391]
1302	NeST–Ada	DreamSim	-0.06804 [-0.07541, -0.06074]
1302	NeST–Ada	CLIP-I	-0.00294 [-0.00857, 0.00249]
1302	NeST–Ada	DISTS	-0.04238 [-0.04529, -0.03948]
1302	WHOLE–Ada	PSNR	+3.44493 [3.28074, 3.61234]
1302	WHOLE–Ada	LPIPS	-0.10599 [-0.11125, -0.10079]
1302	WHOLE–Ada	DreamSim	-0.03769 [-0.04597, -0.02959]
1302	WHOLE–Ada	CLIP-I	-0.02247 [-0.02877, -0.01625]
1302	WHOLE–Ada	DISTS	-0.02772 [-0.03074, -0.02472]
1940	NeST–WHOLE	PSNR	+0.49060 [0.45242, 0.52988]
1940	NeST–WHOLE	LPIPS	-0.01680 [-0.01777, -0.01581]
1940	NeST–WHOLE	DreamSim	-0.01406 [-0.01570, -0.01247]
1940	NeST–WHOLE	CLIP-I	+0.00923 [0.00637, 0.01217]
1940	NeST–WHOLE	DISTS	-0.00870 [-0.00955, -0.00786]
1940	NeST–Ada	PSNR	+2.97773 [2.80999, 3.15216]
1940	NeST–Ada	LPIPS	-0.06414 [-0.06812, -0.06032]
1940	NeST–Ada	DreamSim	-0.00357 [-0.00817, 0.00074]
1940	NeST–Ada	CLIP-I	-0.00487 [-0.00963, -0.00018]
1940	NeST–Ada	DISTS	-0.00907 [-0.01158, -0.00668]
1940	WHOLE–Ada	PSNR	+2.48713 [2.31261, 2.66864]
1940	WHOLE–Ada	LPIPS	-0.04735 [-0.05157, -0.04334]
1940	WHOLE–Ada	DreamSim	+0.01049 [0.00563, 0.01511]
1940	WHOLE–Ada	CLIP-I	-0.01409 [-0.01920, -0.00923]
1940	WHOLE–Ada	DISTS	-0.00037 [-0.00304, 0.00222]
2248	NeST–WHOLE	PSNR	+0.99631 [0.94081, 1.05089]
2248	NeST–WHOLE	LPIPS	-0.03432 [-0.03578, -0.03286]
2248	NeST–WHOLE	DreamSim	-0.03085 [-0.03332, -0.02848]
2248	NeST–WHOLE	CLIP-I	+0.02054 [0.01698, 0.02425]
2248	NeST–WHOLE	DISTS	-0.01648 [-0.01751, -0.01546]
2248	NeST–Ada	PSNR	+1.89003 [1.74362, 2.04058]
2248	NeST–Ada	LPIPS	-0.02977 [-0.03302, -0.02669]
2248	NeST–Ada	DreamSim	+0.00784 [0.00454, 0.01100]
2248	NeST–Ada	CLIP-I	-0.00439 [-0.00907, 0.00014]
2248	NeST–Ada	DISTS	+0.00651 [0.00445, 0.00847]
2248	WHOLE–Ada	PSNR	+0.89372 [0.73796, 1.05249]
2248	WHOLE–Ada	LPIPS	+0.00455 [0.00100, 0.00805]
2248	WHOLE–Ada	DreamSim	+0.03869 [0.03458, 0.04270]
2248	WHOLE–Ada	CLIP-I	-0.02493 [-0.02997, -0.02010]
2248	WHOLE–Ada	DISTS	+0.02299 [0.02077, 0.02515]
2577	NeST–WHOLE	PSNR	+0.88254 [0.80859, 0.95733]
2577	NeST–WHOLE	LPIPS	-0.03295 [-0.03563, -0.03024]
2577	NeST–WHOLE	DreamSim	-0.02698 [-0.02991, -0.02409]
2577	NeST–WHOLE	CLIP-I	+0.01924 [0.01555, 0.02311]
2577	NeST–WHOLE	DISTS	-0.01409 [-0.01534, -0.01283]
2577	NeST–Ada	PSNR	+0.66614 [0.55643, 0.78015]
2577	NeST–Ada	LPIPS	-0.00152 [-0.00390, 0.00071]
2577	NeST–Ada	DreamSim	+0.01779 [0.01549, 0.02007]
2577	NeST–Ada	CLIP-I	-0.00685 [-0.01099, -0.00266]
2577	NeST–Ada	DISTS	+0.02017 [0.01851, 0.02177]
2577	WHOLE–Ada	PSNR	-0.21640 [-0.34068, -0.09134]
2577	WHOLE–Ada	LPIPS	+0.03143 [0.02804, 0.03476]

Bits	Comparison	Metric	Difference [95% CI]
2577	WHOLE–Ada	DreamSim	+0.04477 [0.04100, 0.04860]
2577	WHOLE–Ada	CLIP-I	-0.02608 [-0.03122, -0.02109]
2577	WHOLE–Ada	DISTS	+0.03426 [0.03241, 0.03607]

S4.2 Actual source lengths and endpoint saturation

Bits	Method	Source bits	Padding bits	Full / 500	NULL / 500
620	Ada	547.58	5.42	0	0
1212	Ada	1139.70	5.30	0	0
1302	Ada	1229.54	5.46	0	0
1940	Ada	1866.82	6.18	4	0
2248	Ada	2170.61	10.39	10	0
2577	Ada	2477.89	32.11	75	0
620	NeST	558.28	4.72	0	0
1212	NeST	1150.05	4.95	0	0
1302	NeST	1239.95	5.05	0	0
1940	NeST	1878.36	4.64	0	0
2248	NeST	2186.54	4.46	0	0
2577	NeST	2515.72	4.28	0	0
620	WHOLE	427.80	135.20	0	0
1212	WHOLE	915.86	239.14	0	0
1302	WHOLE	918.95	326.05	0	0
1940	WHOLE	1566.43	316.57	0	0
2248	WHOLE	1599.71	591.29	0	0
2577	WHOLE	1958.14	561.86	0	0

Full means all 10 NeST/WHOLE scales or all 256 Ada tokens. These counts are computed once per source, not once per Ada repeat. Padding after a complete representation is not failure to fit a prefix. Source bits exclude the separately stated information overhead.

S5 Fixed-policy channel threshold

The 10-dB policies of WHOLE, NeST and Ada remain frozen at actual SNRs 8, 8.5, 9, 9.5 and 10 dB. Each cell has 100 sources and 20 new AWGN vectors, hence 2000 actual complete frames and 30 000 frames in total. The source's original repeat-zero payload, waveform, session, counter and scrambler identity are retained; only noise changes. RX uses the true noise variance. Ada's original repeat-zero generation identity remains fixed across noise and SNR.

All 30 000 real receptions and both distance metrics have completed. There are 17 497 accepted outcomes and 12 503 body/profile-parse rejections, with no header rejection, unexpected accepted source input, source-decoding rejection or NULL. No additional image generation is needed: accepted prefixes and frozen generation identities match existing outputs. Scoring required 88 new native gray-reference pairs for each distance; all other pairs were reused after identity verification.

All methods reject all 2000 receptions at 8 dB. At 8.5 dB their rejection rates are 93.05–94.00%, at 9 dB they are 13.50–15.75%, and at 9.5 dB they are 0.10–0.20%. No rejection is observed at 10 dB. Source-cluster intervals retain all 20 noise repetitions within each sampled source. This finite scan describes the tested fixed policies, not low-error-rate operation or general channel robustness.

Method	SNR	Metric	Overall mean [95% CI]
Ada	8	Rejection rate	1.00000 [1.00000, 1.00000]
Ada	8	DISTS	0.76824 [0.76057, 0.77563]
Ada	8	DreamSim	0.93515 [0.92703, 0.94316]

Method	SNR	Metric	Overall mean [95% CI]
Ada	8.5	Rejection rate	0.93500 [0.92200, 0.94700]
Ada	8.5	DISTS	0.72680 [0.71557, 0.73771]
Ada	8.5	DreamSim	0.87996 [0.86721, 0.89279]
Ada	9	Rejection rate	0.13500 [0.11950, 0.15100]
Ada	9	DISTS	0.21603 [0.20456, 0.22794]
Ada	9	DreamSim	0.20653 [0.19141, 0.22233]
Ada	9.5	Rejection rate	0.00100 [0.00000, 0.00250]
Ada	9.5	DISTS	0.13146 [0.12526, 0.13813]
Ada	9.5	DreamSim	0.09424 [0.08616, 0.10328]
Ada	10	Rejection rate	0.00000 [0.00000, 0.00000]
Ada	10	DISTS	0.13079 [0.12463, 0.13742]
Ada	10	DreamSim	0.09341 [0.08534, 0.10239]
NeST	8	Rejection rate	1.00000 [1.00000, 1.00000]
NeST	8	DISTS	0.76824 [0.76057, 0.77563]
NeST	8	DreamSim	0.93515 [0.92703, 0.94316]
NeST	8.5	Rejection rate	0.94000 [0.92950, 0.95000]
NeST	8.5	DISTS	0.73115 [0.72148, 0.74057]
NeST	8.5	DreamSim	0.88560 [0.87443, 0.89682]
NeST	9	Rejection rate	0.14950 [0.13200, 0.16750]
NeST	9	DISTS	0.24294 [0.23094, 0.25502]
NeST	9	DreamSim	0.23428 [0.21806, 0.25085]
NeST	9.5	Rejection rate	0.00100 [0.00000, 0.00250]
NeST	9.5	DISTS	0.15110 [0.14561, 0.15663]
NeST	9.5	DreamSim	0.11231 [0.10429, 0.12089]
NeST	10	Rejection rate	0.00000 [0.00000, 0.00000]
NeST	10	DISTS	0.15047 [0.14502, 0.15601]
NeST	10	DreamSim	0.11149 [0.10345, 0.11997]
WHOLE	8	Rejection rate	1.00000 [1.00000, 1.00000]
WHOLE	8	DISTS	0.76824 [0.76057, 0.77563]
WHOLE	8	DreamSim	0.93515 [0.92703, 0.94316]
WHOLE	8.5	Rejection rate	0.93050 [0.91750, 0.94250]
WHOLE	8.5	DISTS	0.72658 [0.71546, 0.73707]
WHOLE	8.5	DreamSim	0.88020 [0.86695, 0.89269]
WHOLE	9	Rejection rate	0.15750 [0.14300, 0.17200]
WHOLE	9	DISTS	0.26020 [0.25049, 0.27014]
WHOLE	9	DreamSim	0.26497 [0.25173, 0.27887]
WHOLE	9.5	Rejection rate	0.00200 [0.00050, 0.00400]
WHOLE	9.5	DISTS	0.16623 [0.15958, 0.17286]
WHOLE	9.5	DreamSim	0.14037 [0.12846, 0.15272]
WHOLE	10	Rejection rate	0.00000 [0.00000, 0.00000]
WHOLE	10	DISTS	0.16500 [0.15844, 0.17169]
WHOLE	10	DreamSim	0.13874 [0.12689, 0.15117]

Method	SNR	Success / 2000	Cond. DISTS	Cond. DreamSim
Ada	8	0	–	–
Ada	8.5	130	0.12820	0.08738
Ada	9	1730	0.13051	0.09304
Ada	9.5	1998	0.13079	0.09340
Ada	10	2000	0.13079	0.09341
NeST	8	0	–	–
NeST	8.5	120	0.14992	0.11011
NeST	9	1701	0.15061	0.11145
NeST	9.5	1998	0.15047	0.11151
NeST	10	2000	0.15047	0.11149
WHOLE	8	0	–	–

Method	SNR	Success / 2000	Cond. DISTS	Cond. DreamSim
WHOLE	8.5	139	0.16504	0.14297
WHOLE	9	1685	0.16541	0.13993
WHOLE	9.5	1996	0.16503	0.13878
WHOLE	10	2000	0.16500	0.13874

S6 Selected reconstructions

The main figure uses source index 84 (television) and 143 (patterned leather), both at $N = 1024$, 7 dB. They were selected from the complete common500 contact sheets after inspection, using DISTS/DreamSim improvements and visible source fidelity. They are illustrative advantages, not random or representative samples. Original method-specific repeat-zero noise/generation identities remain fixed. Full source IDs, native array hashes, actual prefixes and all five displayed-output metrics are in the figure-source CSVs. The standalone all-method comparison PDF retains the full 256×256 field, including the old Quarter and four additional systems.

S7 Unified processing costs

Timing uses the first 32 common500 IDs, batch one, two warmups and three measurements per source, one actual RTX PRO 6000 and two CPU cores. Original numerical routes are retained; six library compute threads are confined to those two logical CPUs. Model loading, disk I/O, logging and quality scoring are excluded. TX starts with in-memory canonical RGB and ends with actual transmitted symbols. RX starts with in-memory received symbols and ends with canonical RGB. GPU boundary synchronization surrounds each measured span; TX+RX is a processing sum, not radio propagation latency.

The measured operations include tokenizer, condition extraction, CDF/arithmetic coding, finalization/fitting, paid packaging, LDPC/QAM and real receiver PHY, independent source decoding, completion and image decoding. Ada includes ConvNeXt prediction; ARPC includes caption generation; BPG includes its actual search and codec invocation. BPG is CPU-only on the same host and CPU quota; necessary codec serialization uses RAM files. Fresh timing does not substitute cached source tokens or reconstructions. Successful-RX conditional time is reported separately from fast rejection. Means first average within source; median/p95 use measured calls with linear quantiles. The old six-core and component-sum results are not inserted here.

All 16 groups and 32 TX/RX stages completed with actual zero exits and no live owner or child. There are 5120 fresh timing records: 2048 warmups and 3072 measured boundary spans. The 14 previously completed groups were retained; only ARPC and BPG required the explicitly authorized final continuation. Failed setup/measurement attempts remain in the execution/resource record and are excluded from completed-trial latency statistics, rather than converted to zero latency.

ARPC retains its original model runtime and invokes the unchanged qualified CPU PHY through an in-memory socket. Per-frame IPC is included in the measured boundaries; both processes share the same two-CPU affinity. A Python module-name collision was repaired by prebinding the exact standard-library module in the model process and its compiler subprocesses. BPG retains the original codec and search; its child-process CPU-count adapter now matches the registered two-core affinity. Neither repair changes model weights, numerical precision, codec options or transmitted information.

S7.1 All measured groups

Values below are milliseconds. Means average the three measurements within each of 32 sources; median and p95 are calculated from the 96 measured calls. Total pairs the TX and RX spans for the same source and repetition. The success count refers to measured RX calls; conditional RX means exclude rejected receptions and are not substituted for overall RX means.

System	N	SNR	Span	Mean (ms)	Median	p95
WHOLE	1024	7	TX	263.35	262.01	270.94
WHOLE	1024	7	RX	375.26	374.77	379.50

System	N	SNR	Span	Mean (ms)	Median	p95
WHOLE	1024	7	total	638.61	637.56	646.96
NeST	1024	7	TX	362.09	360.53	370.70
NeST	1024	7	RX	390.86	390.64	394.25
NeST	1024	7	total	752.95	751.30	761.83
Ada	1024	7	TX	7546.57	7537.00	7627.76
Ada	1024	7	RX	7873.85	7864.66	7932.79
Ada	1024	7	total	15420.41	15404.73	15537.01
WHOLE	1024	10	TX	213.83	213.37	216.53
WHOLE	1024	10	RX	369.90	371.15	387.29
WHOLE	1024	10	total	583.73	585.23	602.20
NeST	1024	10	TX	248.38	247.09	251.45
NeST	1024	10	RX	411.64	401.40	431.95
NeST	1024	10	total	660.02	649.47	678.57
Ada	1024	10	TX	7795.61	7790.73	7854.45
Ada	1024	10	RX	7626.11	7623.30	7667.14
Ada	1024	10	total	15421.72	15412.42	15498.35
WHOLE	512	10	TX	157.33	157.00	159.30
WHOLE	512	10	RX	325.48	325.10	328.35
WHOLE	512	10	total	482.81	482.32	485.94
NeST	512	10	TX	163.34	163.14	164.79
NeST	512	10	RX	348.89	348.50	352.27
NeST	512	10	total	512.22	511.81	515.45
Ada	512	10	TX	7378.20	7373.40	7423.35
Ada	512	10	RX	7789.73	7785.50	7836.21
Ada	512	10	total	15167.93	15159.94	15239.95
ARPC	1024	10	TX	661.72	653.02	732.59
ARPC	1024	10	RX	797.03	796.61	803.32
ARPC	1024	10	total	1458.76	1448.62	1531.35
Swin	1024	10	TX	20.53	20.47	20.67
Swin	1024	10	RX	17.61	17.60	17.73
Swin	1024	10	total	38.14	38.08	38.41
Latent	1024	10	TX	15.42	15.41	15.47
Latent	1024	10	RX	24.10	24.05	24.21
Latent	1024	10	total	39.51	39.47	39.64
BPG	1024	10	TX	2441.98	2457.79	2827.31
BPG	1024	10	RX	31.71	31.58	33.13
BPG	1024	10	total	2473.69	2489.51	2859.12
Grid Whole	1024	10	TX	183.50	182.95	186.63
Grid Whole	1024	10	RX	378.16	380.12	393.96
Grid Whole	1024	10	total	561.66	569.55	578.15
Grid Quarter	1024	10	TX	214.09	213.20	216.88
Grid Quarter	1024	10	RX	385.21	383.68	388.92
Grid Quarter	1024	10	total	599.30	597.40	609.10
Grid Eighth	1024	10	TX	217.25	216.96	219.00
Grid Eighth	1024	10	RX	388.82	386.04	414.90
Grid Eighth	1024	10	total	606.07	603.35	630.91

System	N	SNR	Success / 96	Conditional RX (ms)
WHOLE	1024	7	96	375.26
NeST	1024	7	96	390.86
Ada	1024	7	96	7873.85
WHOLE	1024	10	96	369.90
NeST	1024	10	96	411.64
Ada	1024	10	96	7626.11
WHOLE	512	10	96	325.48

System	N	SNR	Success / 96	Conditional RX (ms)
NeST	512	10	96	348.89
Ada	512	10	96	7789.73
ARPC	1024	10	96	797.03
Swin	1024	10	96	17.61
Latent	1024	10	96	24.10
BPG	1024	10	96	31.71
Grid Whole	1024	10	96	378.16
Grid Quarter	1024	10	96	385.21
Grid Eighth	1024	10	96	388.82

S7.2 Computation and interpretation

The matched grid comparison uses ceiling $(8, 126)$; its token arm reuses the identical NeST timing group at $N = 1024$, 10 dB. The main WHOLE timing group has its own family target and is not the matched-grid WHOLE control. The timed implementation constructs all candidate streams and scans the complete list for feasibility before selecting the longest one. Its recorded legacy field `length_checks` is the selected endpoint's descending rank, not the actual number of tests. The corrected accounting records 5/20/39/352 candidate tests for matched Whole/Quarter/Eighth/Token, separately from the early-stopping comparisons used by the cached quality control in S3. Both rules select the same endpoint; the timing includes this implementation's full scan.

Fresh source-probability traversals, CDF construction steps, independent finalizations, actual tests and endpoint ranks are in `source_fitting_accounting.csv`, derived from the bound code and per-trial witnesses. Stage-level instrumented model/PHY calls are in `model_call_accounting.csv`; they are not exhaustive operator counts. Counters include warmups and are not interchangeable with latency. No prior source-output cache replaces timed model work.

This software implementation includes actual CDF construction, independent arithmetic termination and real digital decoding. The result is a deployment-cost comparison on this host and these original runtimes, not an optimized-hardware throughput bound. Zero observed failures in a timing cell is not a reliability guarantee. No confidence interval is fabricated from the timing repetitions.

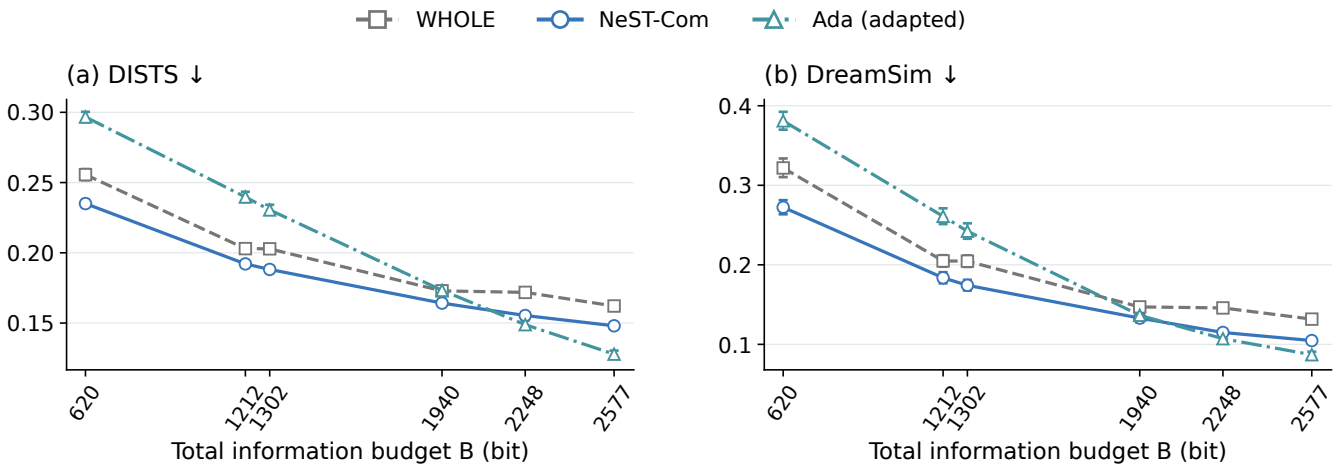


Figure S1: Error-free information-budget comparison on common500. Total bits include all registered information overhead. Numeric budget positions retain their unequal spacing. Points and bars are equal-source means and pointwise 95% source-bootstrap intervals. Ada's three generation repeats are averaged within source; they are not channel repeats. Lower distances are favorable.

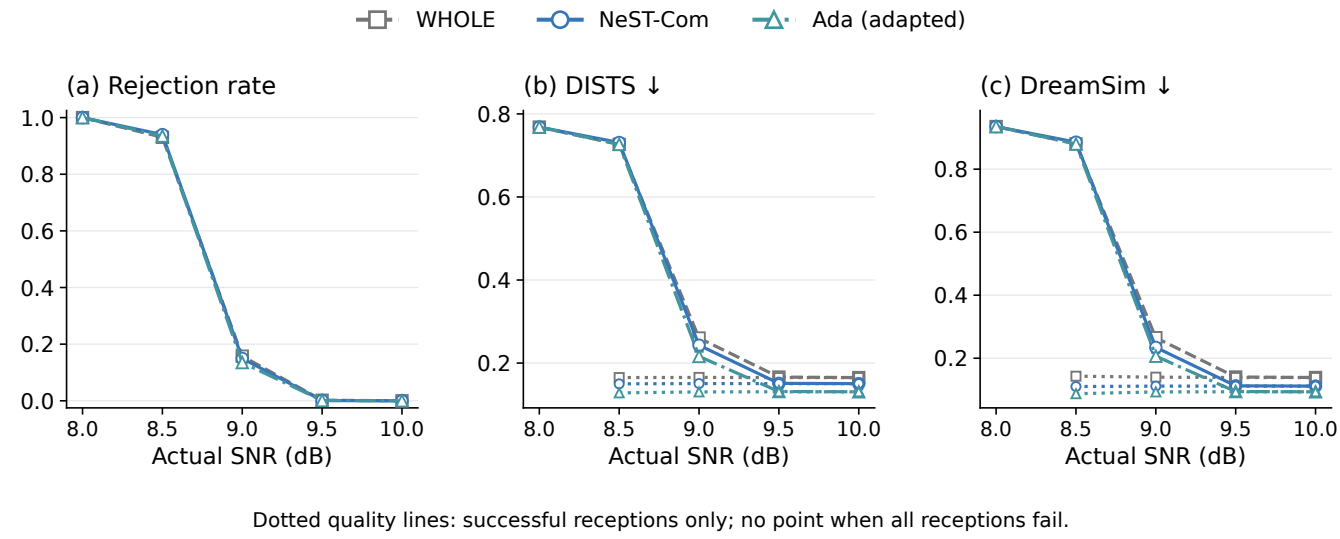


Figure S2: Frozen 10-dB policies evaluated at five actual SNRs. Points and bars retain all receptions and show pointwise 95% source-bootstrap intervals. Dotted quality curves describe successful receptions only and use a changing event denominator; no conditional point exists at 8 dB.